

CABLE TYPE & SIZE	MIL-C-915/49	ACTIVE WIRES	24	CABLE DESIGNATION	R-RN(4)
	UNIT A			UNIT B	
UNIT NUMBER	1			-	
UNIT NAME	TRANSPONDER			ANTENNA CONTROL	
CABLE CONNECTOR	MS3406D28-21S			MS3406D28-21P	
UNIT A TERM NO.	PAIR NO.	COLOR CODE	UNIT B TERM NO.	FUNCTION	
1P8 PIN V	1	BLK	J4 PIN V	REMOTE TURN ON	
1P8 PIN W	1	WHT	J4 PIN W	REMOTE TURN ON RETURN	
1P8 PIN d	1	SHLD	J4 RFI RING		
1P8 PIN c	2	BLK	J4 PIN c	MAIN 15 HZ TRIGGER	
1P8 PIN U	2	WHT	J4 PIN U	MAIN 15 HZ TRIGGER RETURN	
1P8 PIN M	2	SHLD	J4 RFI RING		
1P8 PIN N	3	BLK	J4 PIN N	NOT REQ BY UNIT A (135HZ TRIG)	
1P8 PIN P	3	WHT	J4 PIN P	NOT REQ BY UNIT A (135HZ TRIG RTN)	
1P8 PIN X	3	SHLD	J4 RFI RING		
1P8 PIN e	4	BLK	J4 PIN e	SOFT FAILURE ALARM	
1P8 PIN k	4	WHT	J4 PIN k	SOFT FAILURE ALARM RETURN	
1P8 PIN j	4	SHLD	J4 RFI RING		
1P8 PIN h	5	BLK	J4 PIN h	HARD FAILURE ALARM	
1P8 PIN b	5	WHT	J4 PIN b	HARD FAILURE ALARM RETURN	
1P8 PIN T	5	SHLD	J4 RFI RING		
1P8 PIN L	6	BLK	J4 PIN L	BEACON READY, MSB	
1P8 PIN F	6	WHT	J4 PIN F	1.4 VDC BIAS	
1P8 PIN G	6	SHLD	J4 RFI RING		
1P8 PIN H	7	BLK	J4 PIN H	Y MODE	
1P8 PIN J	7	WHT	J4 PIN J	1.4 VDC BIAS	
1P8 PIN R	7	SHLD	J4 RFI RING		
1P8 PIN Z	8	BLK	J4 PIN Z	ANT BAND SWITCH, MSB	
1P8 PIN f	8	WHT	J4 PIN f	1.4 VDC BIAS	
1P8 PIN m	8	SHLD	J4 RFI RING		
1P8 PIN s	9	BLK	J4 PIN s	BEACON READY	
1P8 PIN r	9	WHT	J4 PIN r	1.4 VDC BIAS	
1P8 PIN p	9	SHLD	J4 RFI RING		
1P8 PIN A	10	BLK	J4 PIN A	ANT BAND SWITCH	
1P8 PIN B	10	WHT	J4 PIN B	1.4 VDC BIAS	
1P8 PIN C	10	SHLD	J4 RFI RING		
1P8 PIN n	11	BLK	J4 PIN n*	SIGNAL GROUND	
1P8 PIN g*	11	WHT	J4 PIN g	CHASSIS GROUND	
1P8 PIN a	11	SHLD	J4 RFI RING		
1P8 PIN S	12	BLK	J4 PIN S	UNUSED	
1P8 PIN K	12	WHT	J4 PIN K	SIGNAL GROUND	

NOTES: *Use EMI backshell, part number: 013-011-001 with connector, part number: 10-21468-21S

n* Tie back wire (do not connect to pin n)

g* Tie to RFI Ring

	SIZE	CAGE CODE	DRAWING NO.	REV
	A	55910	WL-0231674	-
SCALE: NONE		SHEET 2		

CABLE TYPE & SIZE	MIL-C-915/49	ACTIVE WIRES	24	CABLE DESIGNATION	R-RN(4) CONT.
	UNIT A			UNIT B	
UNIT NUMBER	1			-	
UNIT NAME	TRANSPONDER			ANTENNA CONTROL	
CABLE CONNECTOR	MS3406D28-21S			MS3406D28-21P	
UNIT A TERM NO.	PAIR NO	COLOR CODE	UNIT B TERM NO.	FUNCTION	
1P8 RFI RING	12	SHLD	J4 RFI RING		
1P8 PIN D	JUMPER			JUMPER 1P8 PIN D TO 1P8 PIN E	
1P8 PIN E	JUMPER			JUMPER 1P8 PIN E TO 1P8 PIN D	
1P8 RFI RING	DRAIN	NOT SPEC		USE 20 AWG STRANDED WIRED TO	
	WIRE			CONNECT ALL SHIELDS TO BACKSHELL	
				OR CHASSIS GROUND FOR 1P8	
	DRAIN	NOT SPEC	J4 RFI RING	USE 20 AWG STRANDED WIRED TO	
	WIRE			CONNECT ALL SHIELDS TO BACKSHELL	
				OR CHASSIS GROUND FOR J4	

CABLE TYPE & SIZE	1 MIL-C-915/45	ACTIVE WIRES	3	CABLE DESIGNATION	R-RN(8)
	UNIT A			UNIT B	
UNIT NUMBER	1			2	
UNIT NAME	TRANSPONDER			CONTROL-INDICATOR	
CABLE CONNECTOR	MS3456W165S-1P W/BS			MS3456W165S-1P W/BS	
UNIT A TERM NO.	PAIR NO	COLOR CODE	UNIT B TERM NO.	FUNCTION	
1P7 G	1	1B	2P2 G	ETHERNET/DATA	
1P7 A	2	1W	2P2 A	ETHERNET/DATA	
1P7 B	3	2B	2P2 B	GROUND	
1P7 C	4	2W	2P2 C	SPARE	
1P7 D	5	3B	2P2 D	SPARE	
1P7 E	6	3W	2P2 E	SPARE	
1P7 F	7		2P2 F	SPARE	
				SIZE	REV
				A	-
				CAGE CODE	
				55910	
				DRAWING NO.	
				WL-0231674	
				SCALE: NONE	SHEET 3

NOTES:

- △₉ COMPLEMENTARY SIGNALS.
△₁₀ COMPLEMENTARY SIGNALS.

EQUIPMENT NOMENCLATURE: TRANSPONDER GROUP, TACAN
UNIT NOMENCLATURE: TRANSPONDER, SS-24/URN-32
UNIT NUMBER: 1

PLUG/JACK	PIN	FUNCTION NAME	I/O	TYPE	SIGNAL PARAMETER	CHARACTERISTICS			SEC
						NORMAL	MIN	MAX	
△ ₉ 1J8	H	Y MODE (Y_MODE_H)	O		APPLICATION	CONTROL SIGNAL			U
					VOLTAGE LOGIC I	+5 VDC	+2.5 VDC	+5.5 VDC	
					VOLTAGE LOGIC O	0.2 VDC	-0.5 VDC	+0.5 VDC	
					CURRENT LOGIC I		0 mA	1 mA	
					CURRENT LOGIC O		0 mA	20 mA	
					PRF				
					RISE TIME				
					FALL TIME				
△ ₉ 1J8	J	ANTENNA DIFF INTF BIAS (ANT_LOGIC_BIAS)	O		APPLICATION	CONTROL SIGNAL			U
					VOLTAGE (BIAS)	+1.4 VDC	+1.2 VDC	+1.6 VDC	
					VOLTAGE LOGIC (N/A)	N/A			
					CURRENT LOGIC I		0 mA	20 mA	
					CURRENT LOGIC O				
					PRF				
					RISE TIME				
					FALL TIME				
1J8	R	Y MODE SHIELD (GND)	O		APPLICATION	SIGNAL SHIELD			U
					VOLTAGE LOGIC I				
					VOLTAGE LOGIC O				
					CURRENT LOGIC I				
					CURRENT LOGIC O				
					PRF				
					RISE TIME				
					FALL TIME				
△ ₁₀ 1J8	Z	ANTENNA BAND SW MSB (ANT_BAND_SW_MSB_H)	O		APPLICATION	CONTROL SIGNAL MSB			U
					VOLTAGE LOGIC I	+5 VDC	+2.5 VDC	+5.5 VDC	
					VOLTAGE LOGIC O	0.2 VDC	-0.5 VDC	+0.5 VDC	
					CURRENT LOGIC I		0 mA	1 mA	
					CURRENT LOGIC O		0 mA	20 mA	
					PRF				
					RISE TIME				
					FALL TIME				
△ ₁₀ 1J8	f	ANTENNA DIFF INTF BIAS (ANT_LOGIC_BIAS)	O		APPLICATION	CONTROL SIGNAL MSB			U
					VOLTAGE LOGIC I	+1.4 VDC	+1.2 VDC	+1.6 VDC	
					VOLTAGE LOGIC O	N/A			
					CURRENT LOGIC I		0 mA	20 mA	
					CURRENT LOGIC O				
					PRF				
					RISE TIME				
					FALL TIME				
1J8	m	ANTENNA BAND SW SHIELD MSB (GND)	I		APPLICATION	SIGNAL SHIELD			U
					VOLTAGE LOGIC I				
					VOLTAGE LOGIC O				
					CURRENT LOGIC I				
					CURRENT LOGIC O				
					PRF				
					RISE TIME				
					FALL TIME				

NOTES:

- △11 COMPLEMENTARY SIGNALS.
△12 COMPLEMENTARY SIGNALS.

EQUIPMENT NOMENCLATURE: TRANSPONDER GROUP, TACAN UNIT NOMENCLATURE: TRANSPONDER, SS-24/URN-32 UNIT NUMBER: 1									
PLUG/JACK	PIN	FUNCTION NAME	I/O	TYPE	SIGNAL PARAMETER	CHARACTERISTICS			SEC
						NORMAL	MIN	MAX	
△11	1J8	L BEACON READY MSB (BEACON_READY_MSB_H)	O		APPLICATION	CONTROL SIGNAL MSB			U
					VOLTAGE LOGIC I	5.0 VDC	2.5 VDC	+5.5 VDC	
					VOLTAGE LOGIC O	0.2 VDC	-0.5 VDC	+0.5 VDC	
					CURRENT LOGIC I		0 mA	1.0 mA	
					CURRENT LOGIC O		0 mA	20 mA	
					PRF				
					RISE TIME				
					FALL TIME				
△11	1J8	F ANTENNA DIFF INTF BIAS (ANT_LOGIC_BIAS)	O		APPLICATION	CONTROL SIGNAL RETURN MSB			U
					VOLTAGE BIAS	+1.4 VDC	+1.2 VDC	+1.6 VDC	
					VOLTAGE LOGIC O	N/A			
					CURRENT LOGIC I		0 mA	20 mA	
					CURRENT LOGIC O				
					PRF				
					RISE TIME				
					FALL TIME				
	1J8	G BEACON READY SHLD MSB (GND)	I		APPLICATION	SIGNAL SHIELD			U
					VOLTAGE LOGIC I				
					VOLTAGE LOGIC O				
					CURRENT LOGIC I				
					CURRENT LOGIC O				
					PRF				
					RISE TIME				
					FALL TIME				
△12	1J8	s BEACON READY (BEACON_READY_H)	O		APPLICATION	CONTROL SIGNAL			U
					VOLTAGE LOGIC I	5.0 VDC	2.5 VDC	+5.5 VDC	
					VOLTAGE LOGIC O	0.2 VDC	-0.5 VDC	+0.5 VDC	
					CURRENT LOGIC I		0 mA	1.0 mA	
					CURRENT LOGIC O		0 mA	20 mA	
					PRF				
					RISE TIME				
					FALL TIME				
△12	1J8	r ANTENNA DIFF INTF BIAS (ANT_LOGIC_BIAS)	O		APPLICATION	CONTROL SIGNAL RETURN MSB			U
					VOLTAGE LOGIC I	+1.4 VDC	+1.2 VDC	+1.6 VDC	
					VOLTAGE LOGIC O	N/A			
					CURRENT LOGIC I		0 mA	20 mA	
					CURRENT LOGIC O				
					PRF				
					RISE TIME				
					FALL TIME				
	1J8	p BEACON READY SHIELD (GND)	I		APPLICATION	SIGNAL SHIELD			U
					VOLTAGE LOGIC I				
					VOLTAGE LOGIC O				
					CURRENT LOGIC I				
					CURRENT LOGIC O				
					PRF				
					RISE TIME				
					FALL TIME				

NOTES:

13 COMPLEMENTARY SIGNALS.

14 CURRENT FLOWS IN CIRCUIT WHEN TRANSPONDER IS CONNECTED TO ANTENNA CONTROL UNIT AND CONTROL UNIT IS IN REMOTE MODE.

EQUIPMENT NOMENCLATURE: TRANSPONDER GROUP, TACAN UNIT NOMENCLATURE: TRANSPONDER, SS-24/URN-32 UNIT NUMBER: 1									
PLUG/JACK	PIN	FUNCTION NAME	I/O	TYPE	SIGNAL PARAMETER	CHARACTERISTICS			SEC
						NORMAL	MIN	MAX	
13	1J8	A			APPLICATION	CONTROL SIGNAL			U
					VOLTAGE LOGIC I	+5 VDC	+2.5 VDC	+5.5 VDC	
					VOLTAGE LOGIC O	0.2 VDC	-0.5 VDC	+0.5 VDC	
					CURRENT LOGIC I		0 mA	1 mA	
					CURRENT LOGIC O		0 mA	20 mA	
					PRF				
					RISE TIME				
					FALL TIME				
13	1J8	B			APPLICATION	CONTROL SIGNAL RETURN			U
					VOLTAGE LOGIC I	+1.4 VDC	+1.2 VDC	+1.6 VDC	
					VOLTAGE LOGIC O	N/A	-0.5 VDC	+0.5 VDC	
					CURRENT LOGIC I		0 mA	20 mA	
					CURRENT LOGIC O				
					PRF				
					RISE TIME				
					FALL TIME				
	1J8	C			APPLICATION	SIGNAL SHIELD			U
					VOLTAGE LOGIC I				
					VOLTAGE LOGIC O				
					CURRENT LOGIC I				
					CURRENT LOGIC O				
					PRF				
					RISE TIME				
					FALL TIME				
14	1J8	V			APPLICATION	CONTROL SIGNAL			U
					VOLTAGE LOGIC I	24.0 VDC	23 VDC	27.5 VDC	
					VOLTAGE LOGIC O	26 VDC	24 VDC	+28 VDC	
					CURRENT LOGIC I	60 mA	20 mA	200 mA	
					CURRENT LOGIC O	10 mA	0 mA	10 mA	
					PRF				
					RISE TIME				
					FALL TIME				
14	1J8	W			APPLICATION	CONTROL SIGNAL RETURN			U
					VOLTAGE LOGIC I				
					VOLTAGE LOGIC O				
					CURRENT LOGIC I				
					CURRENT LOGIC O				
					PRF				
					RISE TIME				
					FALL TIME				
	1J8	d			APPLICATION	SIGNAL SHIELD			U